Materials/Methods: From December 2001 to January 2007, a total of 41 patients with histologically proven glioblastoma treated with 3D-conformal radiotherapy of 64 Gy or higher plus concurrent and adjuvant temozolomide, were analyzed. The median age was 55 years (range, 28-69 years). Total radiation doses were 64-76 Gy (median 70 Gy, fraction size 2 Gy). Gross tumor volume (GTV) was defined as surgical margin or gadolinium enhanced lesion observed in the T1 sequence MRI series. Clinical target volume (CTV) was defined as 1.5 cm margin from GTV. Surrounding edema outside more than 1.5 cm from GTV was not included in order to reduce the risk of late radiation-associated complication. The irradiated dose to CTV was 44 or 46 Gy. Temozolomide was administered during radiotherapy (75 mg/m^2/d from the first to the last day of radiotherapy or 150-200 mg/m^2/d x 5 days, every 3 weeks for 2 cycles) and after radiotherapy (150-300 mg/m^2/d x 5 days, every 3-4 weeks for 4-6 cycles). Recurrences within GTV were considered central; those within CTV and outside CTV were considered marginal and out-field, respectively.

Results: The majority of patients underwent a total or subtotal resection of their tumors (66%). With a median follow-up of 17 months, 25 patients (61%) died 3-39 months (median, 15.8 months) after operation. Ten (24%) patients were alive without disease, and 5 (15%) patients were alive with disease. Their median follow-up times were 24 months (range, 9-60 months) and 16 months (range, 12-36 months). The median progression-free and overall survival times were 14.7 ± 2.6 months and 22.4 months ± 3.6 (3-60). The 1-year and 2-year survival rates were 77% and 41%. The patterns of failure was central in 13 of 30 (44%), marginal in 7 of 30 (23%), out-field in 3 of 30 (10%), and CSF seeding in 7 of 30 (23%). Radiation necrosis observed in the MRI occurred in 18 patients (44%). Fourteen patients had Grade 1, 3 patients had Grade 2, 1 patient had Grade 3. No patient had Grade 4 radiation necrosis.

Conclusions: Dose escalation of radiotherapy was feasible in the setting of concurrent chemoradiotherapy and resulted in improved survival for glioblastoma. Our results deserve further well-designed investigation of radiation dose escalation study for glioblastoma.

Author Disclosure: C. Suh, None; Y. Suh, None; J. Cho, None.

2107 Assessment of Intracranial Control in Patients Treated with Gamma Knife Radiosurgery

C. H. Matthews[1], J. L. Fox[1], S. Batra[1], K. A. Carson[2], D. Rigamonti[3], L. Kleinberg[1]

[1]*Department of Radiation Oncology and Molecular Radiation Sciences: The Johns Hopkins University, Baltimore, MD,* [2]*Department of Epidemiology: Johns Hopkins Bloomberg School of Public Health, Baltimore, MD,* [3]*Department of Neurosurgery: The Johns Hopkins University, Baltimore, MD*

Purpose/Objective(s): The purpose of this study was to assess the local control and time to progression of intracranial metastases treated with gamma knife (GK) and to assess whether number of lesions influenced this outcome.

Materials/Methods: Two hundred thirty-one patients were treated for brain metastases with GK at Johns Hopkins Hospital between 3/03 and 6/07. Sixty-six patients received GK alone after failing WBRT, 115 received RS alone for initial metastases and 50 received GK as a boost to WBRT. One hundred eight patients were women and 123 were men, with a median age of 56. Histology was non-small-cell lung cancer in 34%, breast in 14%, melanoma in 12%, renal cell carcinoma in 6%, and other in 34%. The majority of patients were RPA class 2. Progression was defined as increase in total volume of tumor and/or the presence of new metastases, as determined by follow-up MRI imaging.

Results: The probability of intracranial progression was not statistically different amongst patients receiving GK alone, GK with upfront WBRT and GK for recurrence. Median time to progression was also not different between these groups (6 months, 6.7 months, and 5.9 months, respectively) ($p = 0.19$, log-rank). Within each treatment group, the probability of intracranial progression was not significantly different in patients with <=3 vs. >3 lesions ($p > 0.58$). Regardless of treatment group, there was a significant difference in time to progression in those patients with <=3 vs. >3 metastases (6.7 vs. 5.8 mo) (log-rank .045). When analyzed by treatment group, the time to progression was only significantly longer for patients with <=3 versus >3 lesions within the GK alone group (7.1 months vs. 4.0 months, respectively).

Conclusions: Treatment approach did not predict for likelihood of progression nor time to progression. However, number of metastases >3 did yield a shorter time to progression in general, and specifically in patients treated with GK alone for new disease. Therefore, the addition of WBRT in patients with >3 metastases might prove beneficial in delaying, but not ultimately preventing, the development of progressive intracranial disease.

Author Disclosure: C.H. Matthews, None; J.L. Fox, None; S. Batra, None; K.A. Carson, None; D. Rigamonti, None; L. Kleinberg, None.

2108 Dose De-escalation with Gamma Knife Radiosurgery in the Treatment of Choroidal Melanoma Where Plaque Brachytherapy is Contraindicated: The Tufts Medical Center Experience

J. E. Mignano, M. Chan, C. Schirmer, L. Williams, J. Duker, K. Yao

Tufts Medical Center, Boston, MA

Purpose/Objective(s): Stereotactic radiosurgery (SRS) provides an alternative to plaque brachytherapy in the treatment of choroidal melanoma when brachytherapy is contraindicated. The potential late toxicity to the optic apparatus has limited the number of patients treated with this approach in the past, however, prior reports on the use of SRS have utilized margin doses in excess of 30 Gy. We present our single institutional experience of a dose de-escalation approach with the intent of visual sparing in the treatment of choroidal melanoma in cases where plaque brachytherapy is contraindicated.

Materials/Methods: From 2000 to 2007, 14 patients with choroidal melanoma and a contraindication to plaque brachytherapy were treated with gamma knife SRS at our institution. The most common contraindications for brachytherapy in our series were juxtapapillary location (8/14) and tumor thickness (3/14). Patients were followed clinically after radiosurgery by a single ophthalmologist. Median follow-up time was 21 months.

Results: Mean tumor thickness was 7 mm (range, 2.1 to 12.2 mm) and mean greatest radial dimension was 16 mm (range, 10 to 25 mm). Mean treated tumor volume was 1.13 ± 1.24 mL. The mean marginal tumor dose was 22.1 ± 2.4 Gy (range, 20 to 26 Gy) and